

Kiyoung Om

Graduate Student, Graduate School of Data Science, KAIST, Daejeon, South Korea
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EDUCATION

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea Mar 2024 – Present
Master of Science in Data Science
Graduate School of Data Science
Advisor: Prof. Jinkyoo Park, System Intelligence Lab

Korea University, Seoul, South Korea Mar 2018 – Feb 2024
Bachelor of Science in Industrial Management Engineering GPA: 4.02/4.5, Major GPA: 4.19/4.5
Military Service: Republic of Korea Army Sep 2019 – Feb 2021

PUBLICATIONS

Conference Papers & Workshop Proceedings

- Yun, T.*, **Om, K.***, Lee, J., Yun, S., & Park, J. (2025). Posterior Inference with Diffusion Models for High-dimensional Black-box Optimization (DiBO). *Accepted in ICML 2025 and ICLR 2025 Workshop*. OpenReview (*Equal contribution)

Preprints

- Om, K.***, Sim, K.*, Yun, T.*, Kang, H., & Park, J. (2025). Posterior Inference in Latent Space for Scalable Constrained Black-box Optimization (CiBO). *Submitted to NeurIPS 2025*. arXiv:2507.00480 (*Equal contribution)

RESEARCH EXPERIENCE

Samsung Electronics-KAIST AI Industry-Academia Collaboration Daejeon, South Korea
AI Research Assistant Mar 2025 – Present

- Developing AI-enhanced supply chain management optimization using subgradient lambda prediction.
- Improving optimization speed and efficiency in complex SCM problems through machine learning approaches.

Korea University Solar Cell Lab AI Industry-Academia Collaboration Seoul, South Korea
AI Application Advisor Apr 2023 – Jul 2023

- Developed prediction models for weighted-average reflection of solar cells using panel images (CNN).
- Created experimental date and efficiency prediction system for Perovskite Solar Cells based on weather forecasting.

TEACHING EXPERIENCE

KAIST Industrial & Systems Engineering Daejeon, South Korea
Teaching Assistant - IE343 Statistical Machine Learning Mar 2024 – Jun 2024

- Supported undergraduate students in developing foundation in machine learning concepts.
- Graded assignments and provided detailed feedback on student work.
- Covered topics including supervised/unsupervised learning, regression, classification, and model evaluation.

Korea University Seoul, South Korea
Teaching Assistant - Deep Learning Programming Study Jul 2023 – Aug 2023

- Assisted students with implementing neural networks using PyTorch under Professor Sungbin Lim.
- Guided participants through practical exercises in deep learning.

SKILLS

Languages	Korean (Native), English (Fluent)
Programming	Python, SQL
ML/DL Frameworks	PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, FinRL
Developer Tools	Git, LaTeX, Jupyter Notebook

AWARDS

Academic Excellence Award

Korea University, 2022

Capstone Excellence Award

Korea University, 2022

Topic: Understanding the Lifecycle and Gap Technologies of Lithium-Ion Batteries

Commander's Commendation

Republic of Korea Army, 2020

Speech Contest on Practicing the Core Values of the Army and Showcasing Examples

EXTRACURRICULAR ACTIVITIES

KUBIG (Korea University BIG data research club)

Seoul, South Korea

Regular Member

Jul 2022 – Jul 2023

- Participated in and led study groups on NLP, Computer Vision, and Reinforcement Learning.
- Developed a vehicle hazard detection system using the YOLOv7 architecture.
- Led an algorithmic trading and backtesting research project using the FinRL framework.

RESEARCH INTERESTS

- Generative Models for Optimization and Control
 - Black-box Optimization
 - Offline Reinforcement Learning
- Alignment of Generative Models via Search and Fine-tuning
 - LLM/Diffusion Models